Text-Guided Dual Refinement for Domain Generalized Semantic Segmentation in Remote Sensing

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***Abstract*—Recently, domain generalized remote sensing semantic segmentation (DG-RSSS) methods leverage vision foundation models (VFMs) with a parameter-efficient fine-tuning (PEFT) strategy to achieve remarkable progress. Although VFMs offer robust representations under distribution shifts between different remote sensing scenes, they still have some limitations, including inaccurate segmentation between similar classes and boundary regions. To address these limitations, this work proposes a text-guided dual refinement (TGDR) approach for DG-RSSS, which contains a text-guided discriminative refinement (TDR) module and a text-guided mask features refinement (TMR) module. In particular, first, the proposed TDR module generates class-discriminated features by using class-related learnable tokens and interclass similarity to refine the original features from the frozen backbone, where the tokens are initialized with class texts. Second, the proposed TMR module injects class-specific semantics into the phase component of mask features by using class-related learnable tokens to enhance the correlation between the semantics of classes and scene contents in mask features for refining the prediction of class boundaries in scene contents. Extensive experiments demonstrate that the proposed TGDR approach achieves superior performance across multiple DG-RSSS benchmarks, e.g., achieving 65.6%, 54.0%, 59.0%, 45.2%, 48.8%, and 79.3% mIoU on the Potsdam–to–Vaihingen (P2V), Vaihingen–to–Potsdam (V2P), Rural–to–Urban (R2U), Urban–to–Rural (U2R), Aerial–to–Satellite II (A2S), and Satellite II–to–Aerial (S2A) benchmarks.**

***Index Terms*—Boundary refinement, class discriminability, domain generalized remote sensing semantic segmentation (DG-RSSS), vision foundation models (VFMs).**

Received 30 April 2026; revised 25 June 2026; accepted 11 August 2026. Date of publication 14 August 2026; date of current version 26 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62561030, Grant 62262028, and Grant 62566030; in part by the Natural Science Foundation of Jiangxi Province, China, under Grant 20262BAC240001, Grant 20252BAC200177, and Grant 20242BAB25082; and in part by the Early Career Young Talents Training Project of Jiangxi Province under Grant 20252BEJ730059 and Grant 20252BEJ730046. *(Corresponding author: Yingqiong Peng.)*

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Digital Object Identifier 10.1109/TGRS.2026.3723822

I. Introduction

SEMANTIC segmentation is a fundamental task in the field of computer vision, aiming to assign a class label to each pixel in an image [1], [2], [3]. Remote sensing semantic segmentation (RSSS) is the extension of this task to remote sensing imagery, with wide applications in environmental monitoring [4], [5], urban planning [6], [7], precision agriculture [8], [9], and disaster assessment [10], [11]. In deep learning, models for remote sensing image segmentation are typically trained in a supervised manner on large annotated datasets (source domain) and evaluated on independent, identically distributed test sets (target domain). However, in practical applications, remote sensing images are acquired using diverse sensors, lighting conditions, spatial resolutions, and geographical contexts. This data heterogeneity introduces severe distribution shifts, limiting the generalization of supervised segmentation models to unlabeled target domains and often leading to significant performance degradation. A straightforward solution is to collect and annotate pixel-level images in the target domain, but this scheme is prohibitively expensive in terms of labor and cost.

To address this challenge, unsupervised domain adaptation is proposed for semantic segmentation [12], [13], [14], [15], which uses a labeled source domain and an unlabeled target domain to train the model, thereby improving performance on the target domain. This learning paradigm alleviates the collection of labels on the target domain. In particular, the prerequisite for unsupervised domain adaptation is to know the target domain to be adapted to, which means having data from the target domain is necessary. However, in practice, it might not be possible to collect all the data from the target domain. Therefore, when the model encounters data that does not fall within the distribution of the target domain, its performance will still deteriorate. To further enhance the generalization ability of the model on any unseen distribution, domain generalized semantic segmentation (DGSS) has emerged, where a model is trained only on source domain data, yet is expected to generalize to arbitrary unseen target domains [16], [17], [18].

Benefit from the robust representations of vision foundation models (VFMs) pre-trained on large-scale datasets under distribution shifts [21], [22], [23], [24], many methods leverage parameter-efficient fine-tuning (PEFT) strategies to

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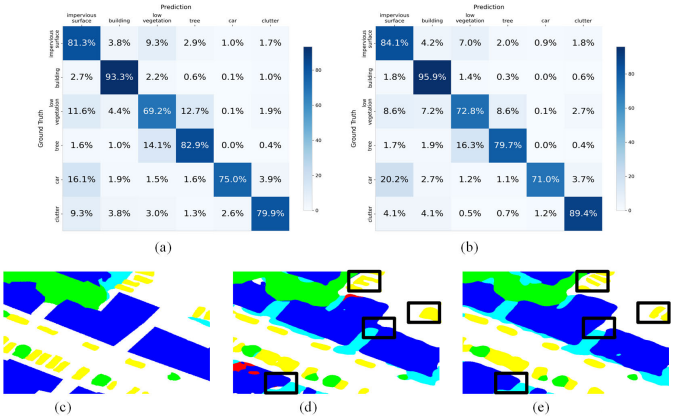


Fig. 1. Phenomena in class confusion and inaccurate segmentation of class boundaries before and after using the proposed TDR and TMR modules. In particular, the baseline is built with the frozen DINOv2 [19] as the backbone and the Mask2Former [20] as the decoder. (a) Base. (b) +TDR. (c) Ground truth. (d) Baseline. (e) +TMR.

fine-tune VFMs for adapting downstream tasks in remote sensing [25], especially for the RSSS task, achieving significant performance improvements. Although VFMs offer robust representations, the class confusion issue still persists due to color and texture similarities among classes. Fig. 1(a) shows the class confusion issue on the Potsdam [26] to Vaihingen [26] (P2V) benchmark, where the segmentation model is built on the frozen DINOv2 and Mask2Former. In addition, as shown in Fig. 1(d), the boundary regions of different class objects in the segmentation results are suboptimal.

To address the aforementioned limitations, this work proposes a text-guided dual refinement (TGDR) approach for domain-generalized RSSS (DG-RSSS), which contains a text-guided discriminative refinement (TDR) module and a text-guided mask features refinement (TMR) module. In particular, for one thing, the proposed TDR module aims to generate class-discriminative features to refine the original features from the frozen backbone. First, we design class-related, learnable tokens, which are initialized by the class-related text embeddings obtained from the CLIP model [27]. Second, the class-related learnable tokens are leveraged to generate class-specific features from the original features. Then, the interclass similarity is calculated between the class-specific features. According to the interclass similarity, the interclass weights are generated to adjust the class-specific features, where high similarities between different classes assign small weights. Finally, the original features and class-specific features are fused by using the cross-attention mechanism to capture the relations between the pixels and classes for generating class-discriminant features, where the original features are set as the query, and the class-specific features are set as the key and value. For another, since the phase components contain more low-frequency components that represent the scene contents, which are relatively stable under the distribution shifts [17], [28], [29], [30], the proposed TMR module injects class-specific semantics into the phase component of mask features, using class-related learnable tokens to enhance the correlation between the semantics of classes and scene contents in mask features for refining the prediction of class boundaries in the scene contents. Extensive experiments demonstrate that the proposed TGDR approach achieves superior performance on multiple DG-RSSS benchmarks.

We summarize the contributions as follows.

1) We propose a TGDR approach for DG-RSSS, which enhances the class discriminability of features and refines the prediction of class boundaries.
2) A TDR module is proposed, which aims to generate class-discriminated features by using class-related learnable tokens and interclass similarity to refine the original features from the frozen backbone.
3) A TMR module is proposed, which aims to enhance the correlation between the semantics of classes and scene contents in mask features by injecting class-specific semantics into the phase component of mask features using class-related, learnable tokens for refining the prediction of class boundaries in the scene contents.
4) Extensive experiments demonstrate that the proposed approach achieves state-of-the-art (SOTA) performance on multiple DG-RSSS benchmarks.

The remainder of this article is organized as follows. Section II reviews related work on DG-RSSS methods and VFM-based DGSS methods in other scenes. Section III details the proposed approach, the TDR module, and the TMR module. Section IV introduces and discusses the benchmarks, experimental settings, and experimental results. Section V states the limitations and future work. Section VI gives the conclusion of this work.

II. Related Work

We briefly review the recent DGSS methods in remote sensing and VFM-based DGSS in other scenes.

A. DGSS Methods in Remote Sensing

DGSS has gradually become a research hotspot in the remote sensing domain, aiming to train models on source domain data that maintain high performance on unseen target domains. Researchers have proposed various approaches in recent years to enhance cross-domain generalization, including category consistency constraints, data augmentation, frequency-based perturbations, and VFM adaptation.

In terms of category consistency, Liang et al. [31] introduced constraints in the feature space to maintain stable semantic discriminability across different remote sensing scenarios, demonstrating significant improvements in adaptation to unseen domains on cross-regional datasets. Regarding data augmentation and sample diversity, Zang et al. [32] proposed a joint style-and-layout synthesis method, generating training samples with diverse styles and layouts to improve model generalization to unseen domains. Luo et al. [33] constructed a large-scale building dataset and combined it with diverse sample augmentation strategies, enabling the model to learn richer spatial structures and perform more robustly in cross-region segmentation tasks. For frequency-based perturbations, Iizuka et al. [34] proposed a frequency-optimal style mixing strategy, adjusting the

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high- and low-frequency components of images during training to introduce source-domain diversity, thereby significantly enhancing model adaptability to unseen domains across multiple land-cover datasets. In the vision foundation adaptation category, Gong et al. [35] proposed the CrossEarth geospatial VFM, integrating remote sensing-specific features for robust cross-regional segmentation. Luo et al. [36] applied VFM fine-tuning for DGSS in remote sensing, combining source domain characteristics with pretrained representations to improve generalization to unseen domains. Wu et al. [37] further explored the open-vocabulary DGSS (OVDG) task, proposing the FreeMix framework to extract domain-agnostic visual features with self-supervised models, integrate them with CLIP for joint recognition of base and novel classes, and enhance multisource domain learning via dataset-aware sampling.

B. VFM-Based DGSS Methods in Other Scenes

With the remarkable progress of VFMs in computer vision, their application in urban-scene DGSS has received increasing attention. Unlike traditional source-domain training, VFMs provide rich general visual representations that help maintain stable performance in unseen domains. To fully leverage the cross-domain capabilities of VFMs, researchers have proposed various fine-tuning strategies.

Jeon et al. [38] proposed the DPMFormer framework, which generates dynamic domain attribute prompts through domain-aware prompt learning and combines them with texture perturbations and consistency constraints to address misalignment between visual and textual semantics caused by fixed prompts. Zhao et al. [39] introduced FisherTune, using a domain-related Fisher information matrix (DR-FIM) to evaluate VFM parameter sensitivity across tasks and domains, selectively updating parameters to enhance cross-domain segmentation robustness while preserving model generalization. Bi et al. [17] proposed FADA, decomposing features into low-frequency (content) and high-frequency (style) components via the Haar wavelet transform, with frequency-domain tokens handling each branch separately to stabilize scene content and mitigate domain style effects. Furthermore, based on the Haar wavelet transform, Bi et al. [40] proposed an efficient frequency learning scheme for diffusion models, which aims to learn structural priors in low-frequency components from the source domain and learn target-related details in high-frequency components from the unseen target domain for improving the generalization ability of Diffusion models in unseen domains. Yi et al. [29] introduced SET, using the Fourier transform to decompose VFM features into phase (content) and amplitude (style) components, which are processed with learnable tokens and attention-based optimization to bridge the gap between style-influenced representations and static tokens. Zhang and Tan [41] proposed the MFuser framework, fusing VFMs with vision-language models (VLMs) through a Mamba architecture, employing collaborative fine-tuning and image-prior-enhanced text embeddings to improve cross-domain segmentation generalization. Yun et al. [42] introduced SoMA, performing singular value decomposition on VFM parameters and selectively fine-tuning minor components to strengthen domain-invariant features and improve cross-domain performance. Chen et al. [43] proposed DepthForge, which integrates VFM visual features with depth information and introduces depth-aware tokens to enforce geometric consistency, thereby enhancing generalization and spatial robustness in unseen domains. Bi et al. [44] proposed a decoupled feature-as-a-query framework, which aims to use the channelwise decoupled features as queries with a restricted isometry instance whitening transform to learn domain generalized medical image representations for multiple medical domain generalization tasks, e.g., whole slide classification, DR grading, and fundus segmentation.

Although these methods achieve promising cross-domain generalization, they still face some limitations: features are not discriminatively calibrated for confusing categories, and the segmentation of boundary regions is suboptimal. To overcome these limitations, this work proposes a TGDR approach for DG-RSSS, which contains a TDR module and a TMR module. The former aims to improve the class discriminability of features, and the latter aims to refine the prediction of class boundaries. In Section III, the proposed TGDR approach is introduced in detail.

C. Compared With Related Works

Recently, some studies on vision–language remote sensing segmentation also rely on text embeddings to optimize the feature learning. For example, Li et al. [45] proposed an RSKT-Seg framework for open-vocabulary remote sensing image segmentation, which contains a multidirectional cost map aggregation module, an efficient cost map fusion transformer, and a remote sensing knowledge transfer module, to learn rotation-invariant features, model semantic relations, and inject pretrained knowledge by using visual and text representations from pretrained CLIP and DINO models. Zhang et al. [46] proposed a hierarchical textual-visual guidance (HTVG) method for referring remote sensing image segmentation, which contains a textual semantic parsing module to transform complex referring expressions into hierarchical attributes encompassing category recognition, spatial constraints, relational semantics, and intrinsic properties for providing explicit cues. Jeon et al. [38] proposed a domain-aware prompt-driven masked transformer (DPMFormer) for DGSS, which contains a domain-aware prompt learning module to achieve semantic alignment between visual and textual cues.

Unlike the recent works above, first, the task addressed by the proposed TGDR approach differs from theirs. The proposed TGDR approach focuses on the DGSS in remote sensing, while RSKT-Seg [45] focuses on open-vocabulary semantic segmentation in remote sensing, HTVG [46] focuses on referring segmentation in remote sensing, and DPMFormer [38] focuses on the DGSS in urban sensing. Second, the proposed TGDR approach uses class-related text embeddings to initialize a set of learnable tokens and then optimizes these tokens to enhance the class discriminability of features and refine the prediction of class boundaries, while RSKT-Seg [45] optimizes the text encoder of the CLIP model to generate class-related text embeddings for enhancing the class

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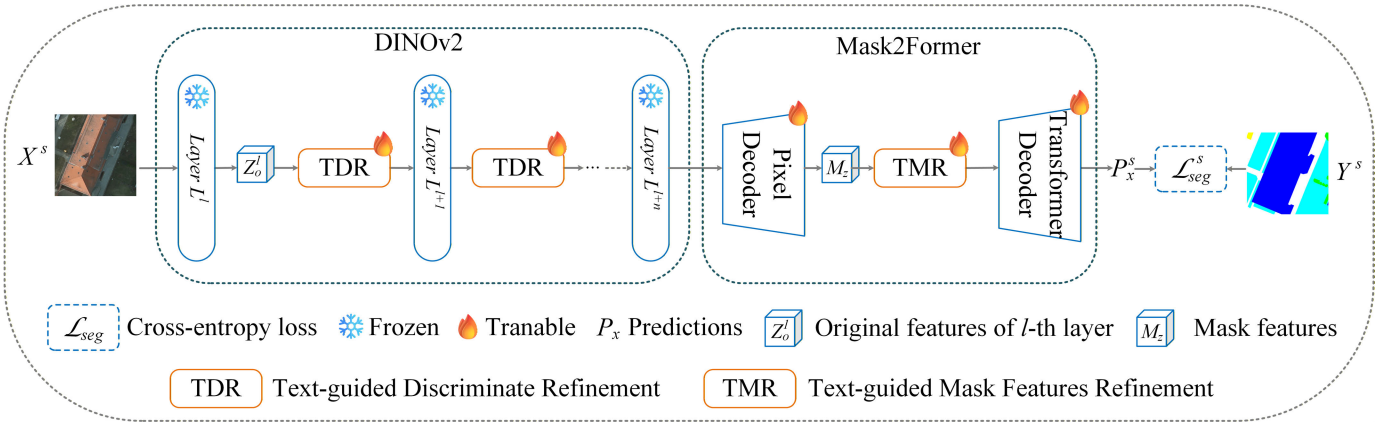


Fig. 2. Pipeline of the proposed TGDR approach.

discriminability. In addition, HTVG [46] transforms the text prompts into hierarchical attributes, which aims to achieve visual grounding rather than mining class-specific semantics from text prompts. DPMFormer [38] aims to use domain-aware text prompts to generate diverse samples rather than to learn class-specific semantics from the text prompts.

III. Approach

A. Overview

The proposed TGDR approach is shown in Fig. 2, which consists of the frozen backbone of the VFMs (e.g., DINOv2 [19]), the proposed TDR module, and the proposed TMR module.

In particular, first, in the frozen backbone of DINOv2, the source domain images (X^s) are fed into the frozen backbone to extract the original features Z_o^l, where l indicates the lth layer. Second, the features Z_o^l are set as the input of the TDR module, which aims to generate the class-specific features Z_c^l to refine the original features Z_o^l for obtaining the final features Z_f^l. In particular, the final features Z_f^l are viewed as the inputs of the next layer. Second, in the Mask2Former, the mask features M_z are generated by feeding the final features Z_f^l into the pixel decoder, and the mask features M_z are fed into the proposed TMR module, which aims to enhance the correlation between the semantics of classes and scene contents in mask features by injecting class-specific semantics to the phase component of mask features using class-related learnable tokens for refining the prediction of class boundaries in scene contents. In particular, the details of the proposed TDR and TMR modules are introduced in the following sections. After refining the mask features M_z, the Transformer decoder is leveraged to generate the class scores of each queried proposal and predicted mask by decoding the refined mask features $\hat{M}_z$. The total loss function of Mask2Former is denoted as follows:

$$\mathcal{L}_{seg}^s = \lambda_{mask}\mathcal{L}_{mask} + \lambda_{dice}\mathcal{L}_{dice} + \lambda_{cls}\mathcal{L}_{cls} \tag{1}$$

where $\mathcal{L}_{mask}$, $\mathcal{L}_{dice}$, and $\mathcal{L}_{cls}$ denote the cross-entropy loss of the predicted mask, dice loss of the predicted mask, and classification loss for each queried proposal. In particular, the hyperparameters λ_{mask}, λ_{dice}, and λ_{cls} are set to 5.0, 5.0, and 2.0, respectively.

B. TDR Module

Although VFMs offer robust representations, the class confusion issue still persists due to color and texture similarities among classes. To address the class confusion issue, we propose the TDR module shown in Fig. 3, which aims to generate class-discriminated features by using class-related learnable tokens and interclass similarity to refine the original features from the frozen backbone.

In particular, first, a set of class-related learnable tokens $T \in \mathcal{R}^{L\times C\times M}$ is proposed, where L, C, and M denote the number of layers in the frozen backbone, the number of tokens, and the number of vectors in each token. In particular, C equals the number of classes, and the class-related learnable tokens T are initialized by the class-related text embeddings obtained from the CLIP model [27], where the prompts consist of a simple sentence with a class text, e.g., "a remote sensing photograph of [cls]." Thus, the semantics of each class are contained in the class-related learnable tokens $T^l \in \mathcal{R}^{C\times M}$, which aim to obtain the class-specific features $Z_c^l \in \mathcal{R}^{B\times C\times N}$ of each layer, where l denotes the lth layer. The class-specific features $Z_c^l \in \mathcal{R}^{B\times C\times N}$ can be denoted as follows:

$$Z_c^l = T^l \times Z_o^l \tag{2}$$

where $Z_o^l \in \mathcal{R}^{B\times M\times N}$ denotes the original features of each layer in the frozen backbone. B, M, and N denote the number of batch size, the number of channels, and the number of vectors in each channel.

Second, the interclass similarity $S_c^l \in \mathcal{R}^{B\times C\times C}$ is calculated between the class-specific features, which can be denoted as follows:

$$S_c^l = Z_c^l \times \left(Z_c^l\right)^T. \tag{3}$$

According to the interclass similarity S_c^l, the interclass weights W_s^l are generated, which can be denoted as follows:

$$W_s^l = \mathbf{1} - \mathcal{N}\left(S_c^l\right) + I \tag{4}$$

where I, $\mathcal{N}(\cdot)$, and $\mathbf{1}$ denote the identity matrix, the normalization operation, and the matrix with all one value, respectively.

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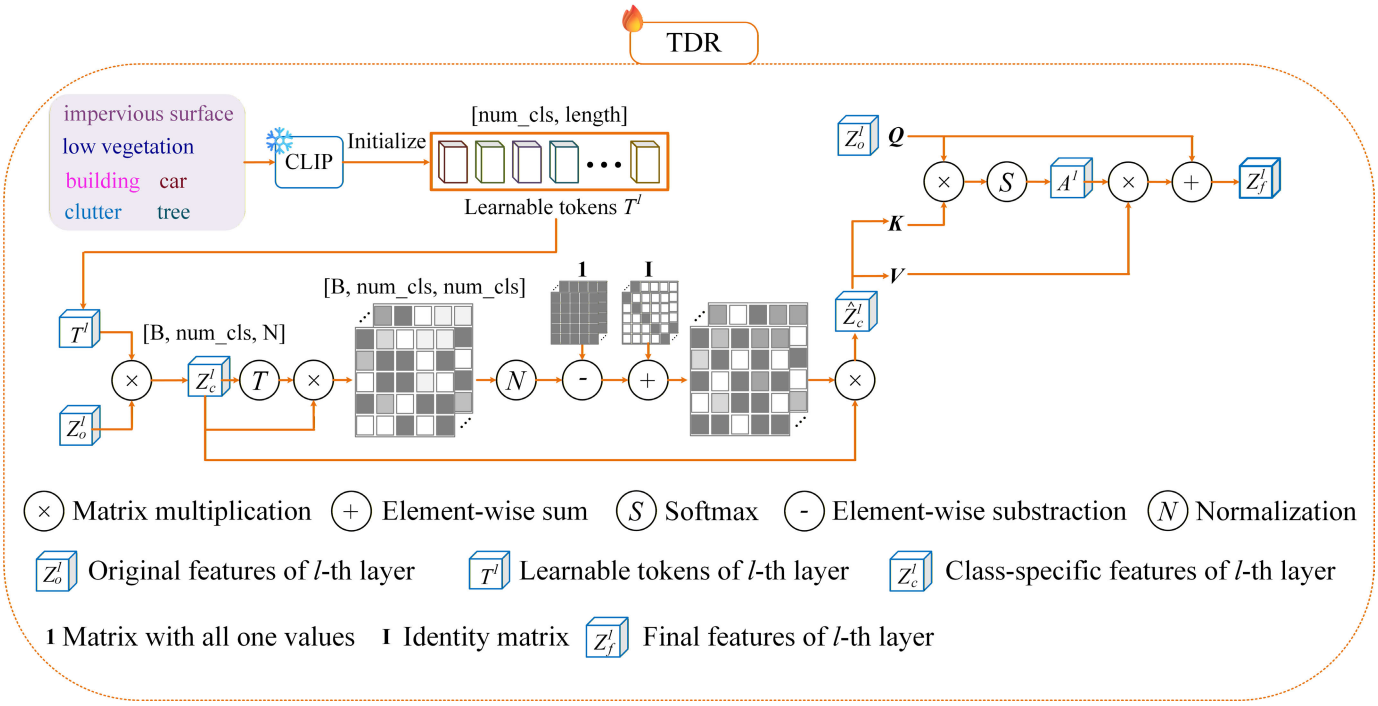


Fig. 3. Pipeline of the proposed TDR module.

In particular, the high similarities between different classes in the inter-class similarity S_c^l assign small weights in the interclass weights W_s^l.

Then, the interclass weights W_s^l are utilized to calibrate the class-specific features, which can be denoted as follows:

$$\hat{Z}_c^l = W_s^l \times Z_c^l \tag{5}$$

where $\hat{Z}_c^l$ denotes the calibrated class-specific features.

Finally, the original features $Z_o^l \in \mathcal{R}^{B\times M\times N}$ and calibrated class-specific features $\hat{Z}_c^l \in \mathcal{R}^{B\times C\times N}$ are fused by using the cross-attention mechanism to capture the relations between the pixels and classes for generating final class-discriminated features Z_f^l, where the original features are set as the query and the class-specific features are set as the keys and values. Thus, the final class-discriminated features Z_f^l can be denoted as follows:

$$Z_f^l = Z_o^l + \left(S\left(\frac{Z_o^l \times \left(\hat{Z}_c^l\right)^T}{\sqrt{d_N}}\right) \times \hat{Z}_c^l\right) \tag{6}$$

where S denotes the Softmax operation. d_N denotes the value of N dimensionality. The final class-discriminant features Z_f^l represent the aggregation of semantic information from pixels to each class.

C. TMR Module

Although VFMs offer robust representations, and the proposed TDR module shown in Fig. 4 further enhances the class discriminability of features from VFMs, as shown in Fig. 1(d), the boundary regions of different class objects in segmentation results are suboptimal. To address this limitation, we propose the TMR module, which aims to enhance the correlation between the semantics of classes and the scene content in mask features by injecting class-specific semantics to the phase component of mask features using class-related learnable tokens for refining the prediction of class boundaries in scene content.

In particular, first, as shown in Fig. 5, because phase components contain more low-frequency components that represent the scene contents, which are relatively stable under the distribution shifts [17], [28], [29], [30], the mask features $M_z \in \mathcal{R}^{B\times K\times H\times W}$ are projected into the frequency space by using 2-D fast Fourier transform in the channel dimensionality K, which can be denoted as follows:

$$M_f(x,y) = \sum_{h=1}^{H}\sum_{w=1}^{W} M_z(h,w)\, e^{-2j\pi\left(x\frac{h}{H}+y\frac{w}{W}\right)} \tag{7}$$

where M_f denotes the frequency representation, which can be divided into two components, that is, the real component M_f^{real} and the imaginary component M_f^{img}, respectively, denoted as follows:

$$M_f = M_f^{real} + iM_f^{img}. \tag{8}$$

According to the real component M_f^{real} and the imaginary component M_f^{img}, the amplitude and phase of M_f can be obtained by using the spectral decomposition, which are denoted as follows:

$$M_f^{amp} = \sqrt{\left(M_f^{real}\right)^2 + \left(M_f^{img}\right)^2} \tag{9}$$

$$M_f^{pha} = \arctan\left(\frac{M_f^{img}}{M_f^{real}}\right) \tag{10}$$

where M_f^{amp} and M_f^{pha} are the amplitude component and phase component of M_f.

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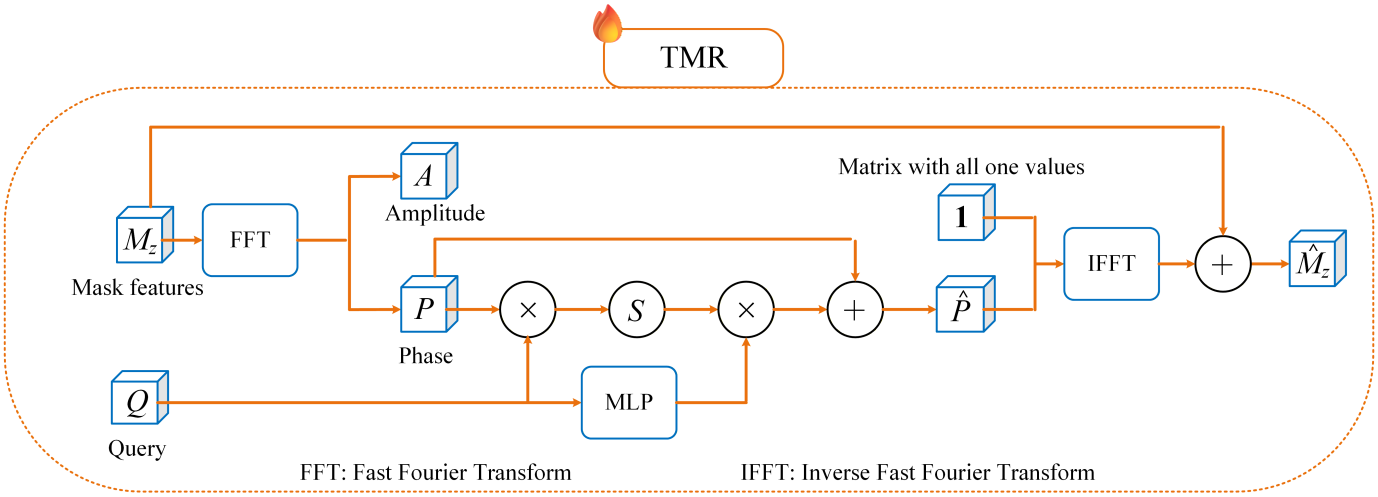


Fig. 4. Pipeline of the proposed TMR module.

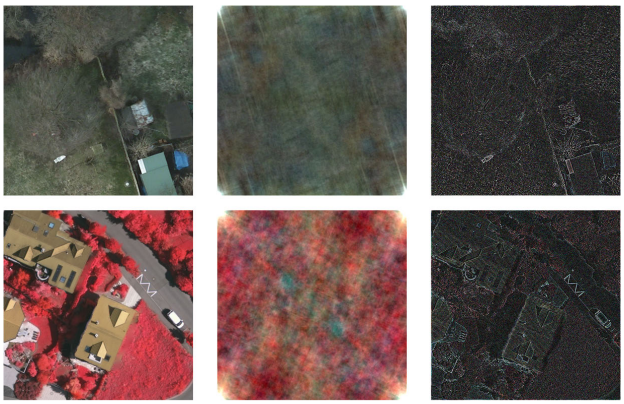


Fig. 5. Examples of the amplitude-only and phase-only reconstruction. (a) Original images (RGB). (b) Only amplitude components are utilized to reconstruct images, where the phase component is set as a constant. (c) Only phase components are utilized to reconstruct images, where the amplitude component is set as a constant.

Second, the class-specific semantics are injected to the phase component of mask features $M_f^{pha} \in \mathcal{R}^{B\times K\times H\times W}$ by using class-related query $Q \in \mathcal{R}^{B\times C\times K}$ to enhance the correlation between the semantics of classes and scene contents in mask features for refining the prediction of class boundaries in scene contents. The process can be denoted as follows:

$$\hat{M}_f^{pha} = M_f^{pha} + \left(MLP(Q)^T \times S\left(\frac{Q \times M_f^{pha}}{\sqrt{d_K}} \right) \right) \tag{11}$$

where $\hat{M}_f^{pha}$ is the refine phase component. MLP denotes a multilayer perceptron. Motivated by Rein [21], Q is generated from the class-related learnable tokens T, which can be denoted as follows:

$$Q = Concat\left(\left[Q_{max}, Q_{avg}, Q^L \right] \right) \tag{12}$$

where Q_{max} is denoted as $\max Q^L$, Q_{avg} is denoted as $1/L \sum_{l=1}^{L} Q^l$, and Q^L is denoted as $Q^L = T^L \times W_{Q^L} + b_{Q^L}$. The W_{Q^L} and b_{Q^L} are the weights and biases of the linear transformation projection.

Then, the M_f can be rerepresented as follows:

$$\hat{M}_f = \mathbf{1} \cos\left(\hat{M}_f^{pha} \right) + i\mathbf{1} \sin\left(\hat{M}_f^{pha} \right) \tag{13}$$

where $\mathbf{1}$ denotes the vector with all one values, which is used to replace the M_f^{amp} to enhance the scene contents for the mask features. $\hat{M}_f$ denotes the refined frequency representation, which is transformed to the spatial space by using the Inverse Fast Fourier Transform. The process can be denoted as follows:

$$M_{z_f}(h,w) = \frac{1}{HW} \sum_{x=1}^{H} \sum_{y=1}^{W} \hat{M}_f(x,y)\, e^{2j\pi\left(x\frac{h}{H} + y\frac{w}{W}\right)}. \tag{14}$$

Finally, the refine mask features $\hat{M}_z$ is obtained by fusing the M_z and M_{z_f}, which can be denoted as follows:

$$\hat{M}_z = M_z + M_{z_f}. \tag{15}$$

IV. Experiments

A. Benchmarks and Datasets

The proposed TGDR approach is evaluated on the DG-RSSS benchmarks. In particular, the DG-RSSS benchmarks include Potsdam–to–Vaihingen (P2V), Vaihingen–to–Potsdam (V2P), Rural–to–Urban (R2U), and Urban–to–Rural (U2R). In particular, following previous work [30], for the Potsdam–and–Vaihingen datasets, we use the RGB band for all experiments. Moreover, following the experimental settings in the recent work [48], the proposed TGDR approach is further evaluated on the Aerial–to–Satellite II (A2S) and Satellite II–to–Aerial (S2A) benchmarks contained in the WHU dataset [49]. The details of the datasets in these benchmarks are as follows.

1) Potsdam [26] dataset includes 38 very high resolution (VHR) True Orthophotos (TOPs) images with 6000 × 6000 pixels.

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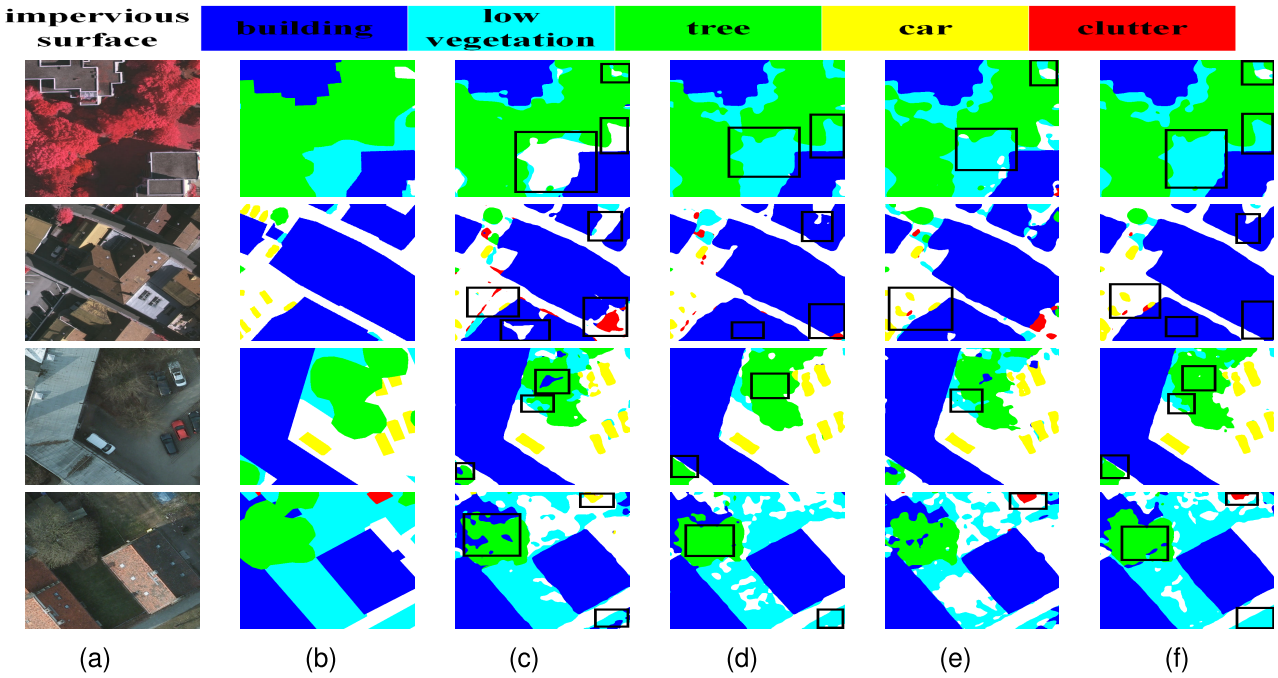


Fig. 6. Visualizations of ablation experiments on the P2V and V2P benchmarks [26]. (a) Input. (b) GT. (c) Base. (d) TDR. (e) TMR. (f) Our.

2) Vaihingen [26] dataset contains 33 VHR TOP images with an average size of 2494 × 2064 pixels.
3) The R2U and U2R benchmarks are contained in the LoveDA dataset [47], which consists of 4190 images with a resolution of 1024 × 1024 pixels, collected from both urban and rural areas.
4) The A2S and S2A benchmarks are contained in the WHU dataset [49], which focuses on the building extraction task, where the Aerial dataset consists of 5772 and 2416 images for training and testing, and the Satellite II dataset consists of 13 662 and 3762 images for training and testing.

B. Implementation Details

1) Baseline: The frozen backbone of DINOv2 [19] and the Mask2Former [20] decoder form the baseline. Moreover, Earth-Adapter [30] is also viewed as another baseline, where the proposed TGDR approach is applied to Rein [21], FADA [17], SET [29], Earth-Adapter [30], and SoMA [42].

2) Training Details: The proposed TGDR approach is trained on two NVIDIA RTX 3090 GPUs with 40 000 iterations. The batch size is set to 4. We use the AdamW optimizer with the initial learning rate of 1×10^{-4} to optimize the proposed TDR module, TMR module, and the decoder. The mIoU is utilized to measure the performance of the proposed TGDR approach.

C. Ablation Experiments

The contribution of each component in the proposed TGDR approach based on different recent methods, including Rein [21], FADA [17], SET [29], Earth-Adapter [30], and SoMA [42], is analyzed on the P2V, V2P, R2U, and U2R benchmarks, which is shown in Table I. Moreover, Tables II and III, Figs. 6 and 7 show the quantitative and qualitative experiments on the P2V, V2P, R2U, and U2R benchmarks to further

TABLE I
ABLATION EXPERIMENTS OF THE PROPOSED TGDR APPROACH BASED ON DIFFERENT RECENT METHODS ON THE P2V, V2P, R2U, AND U2R BENCHMARKS [26], [47]

Methods	TDR	TMR	Average	P2V	V2P	R2U	U2R
Baseline [19]	-	-	53.1	62.2	50.5	55.9	43.9
Ours	√	-	54.9	63.9	**53.6**	58.0	44.2
	-	√	53.9	62.5	51.4	57.5	44.1
	√	√	**55.1**	**64.5**	53.3	**58.1**	**44.6**
Rein [21]	-	-	53.0	62.0	52.3	56.2	41.3
Ours	√	-	54.5	63.3	54.3	57.2	43.3
	-	√	54.2	62.9	54.1	56.9	42.8
	√	√	**55.0**	**63.7**	**54.7**	**57.6**	**44.0**
FADA [17]	-	-	53.5	60.6	52.4	57.0	43.9
Ours	√	-	54.2	61.7	53.1	57.6	44.2
	-	√	54.5	62.5	53.3	58.0	44.3
	√	√	**55.1**	**63.5**	**53.9**	**58.3**	**44.6**
SET [29]	-	-	52.2	60.4	49.7	56.5	42.2
Ours	√	-	53.5	62.2	51.2	57.1	43.4
	-	√	53.3	61.8	50.5	57.3	43.7
	√	√	**54.2**	**63.1**	**51.8**	**57.8**	**44.1**
Earth-Adapter [30]	-	-	54.7	61.6	54.5	58.3	44.4
Ours	√	-	55.3	62.5	**54.7**	58.5	45.5
	-	√	54.9	62.2	54.0	58.4	45.1
	√	√	**55.5**	**63.1**	54.2	**58.8**	**45.8**
SoMA [42]	-	-	54.1	62.3	52.8	58.1	43.3
Ours	√	-	55.2	64.5	53.3	58.4	44.6
	-	√	55.3	65.0	53.4	58.5	44.2
	√	√	**56.0**	**65.6**	**54.0**	**59.0**	**45.2**

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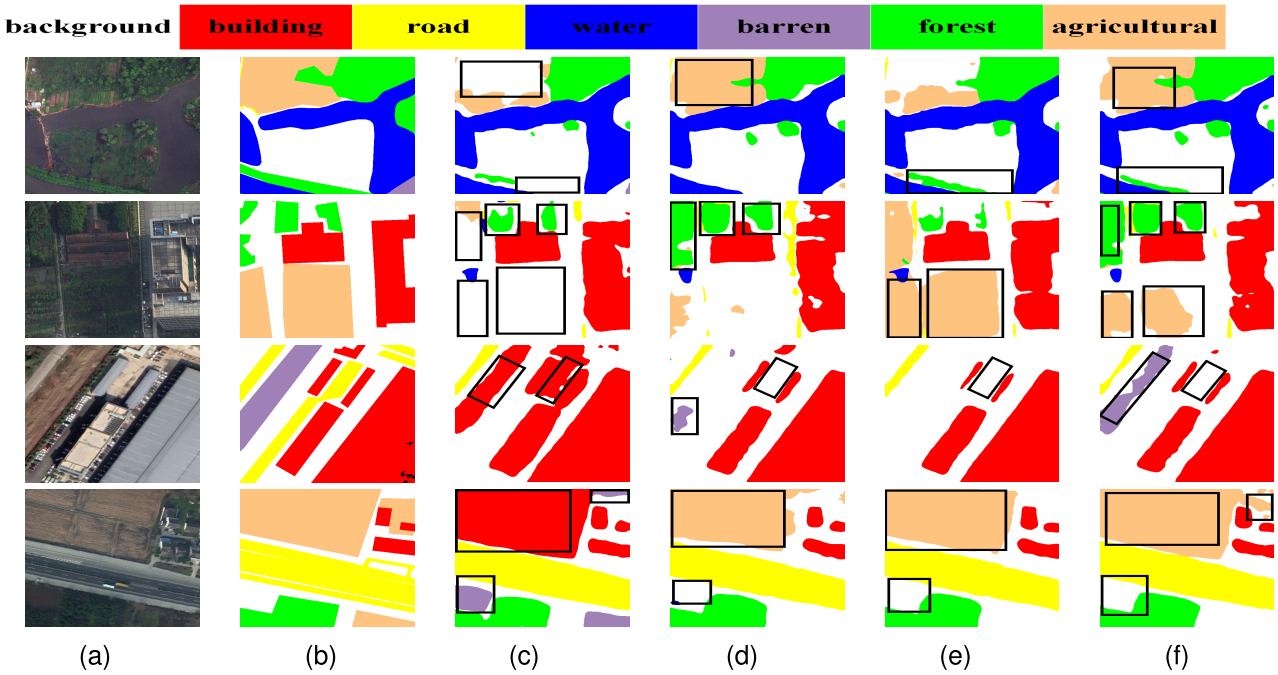


Fig. 7. Visualizations of ablation experiments on the R2U and U2R benchmarks [47]. (a) Input. (b) GT. (c) Base. (d) TDR. (e) TMR. (f) Our.

TABLE II

ABLATION EXPERIMENTS OF THE PROPOSED TDR MODULE ON THE P2V, V2P, R2U, AND U2R BENCHMARKS [26], [47]. IN PARTICULAR, THE "W/O W_s" INDICATES THAT DIRECTLY USING THE INTER-CLASS SIMILARITY S_c TO CALIBRATE THE CLASS-SPECIFIC FEATURES Z_c. THE "W/O S_c" INDICATES THAT DIRECTLY USING THE CLASS-SPECIFIC FEATURES Z_c TO REFINE THE ORIGINAL FEATURES Z_o TO GENERATE THE FINAL CLASS-DISCRIMINATED FEATURES Z_f

Methods	Average	P2V	V2P	R2U	U2R
TDR	54.9	63.9	53.6	58.0	44.2
w/o W_s	54.3	63.5	53.1	56.8	43.8
w/o S_c	53.9	63.1	52.5	56.4	43.5

TABLE III

ABLATION EXPERIMENTS OF THE PROPOSED TMR MODULE ON THE P2V, V2P, R2U, AND U2R BENCHMARKS [26], [47]. IN PARTICULAR, THE "W/ REFINE M_f^{amp}" DENOTES THAT THE M_f^{pha} IS REPLACED WITH M_f^{amp} IN 11. THE "W/O $\mathbf{1}$" DENOTES THAT THE $\mathbf{1}$ IS REPLACED WITH M_f^{amp} IN 13

Methods	Average	P2V	V2P	R2U	U2R
TMR	53.9	62.5	51.4	57.5	44.1
w/ refine M_f^{amp}	53.2	62.1	51.0	56.2	43.5
w/o $\mathbf{1}$	53.4	62.3	51.1	56.4	43.8

verify the effectiveness of each component in the proposed TGDR approach, respectively. In particular, we use the average performance of four benchmarks to evaluate the effectiveness of each proposed module.

As shown in Table I, when adding the proposed TDR and TMR to the baseline and recent works, the performance increased in succession, except for adding the proposed TMR module to the Earth-Adapter [30] on the V2P benchmark. For example, compared with the baseline, the proposed TDR and TMR modules achieve improvements of 1.8% and 0.8% mIoU in the average performance. Compared with SoMA [42], the proposed TDR and TMR modules achieve improvements of 1.1% and 1.2% mIoU in the average performance. After combining these two modules, the average performance could be further increased. These successive performance improvements demonstrate the effectiveness of each component in the proposed TGDR approach.

However, when adding the proposed TMR module to Earth-Adapter [30], the performance decreased by 0.5%. In particular, the learned features in the Earth-Adapter [30], which are used to generate the mask features, are combined into three parts, including spatial features, high-frequency features, and low-frequency features, by using three learnable weights. Thus, for the learned features in the Earth-Adapter, the spatial, low-frequency, and high-frequency information is combined with suitable weights, which could be used to generate better mask features. It means all spatial, low-frequency, and high-frequency information contained in the mask features in the Earth-Adapter [30] may be beneficial for the semantic segmentation task. When the proposed TMR module further enhances the correlation between the semantics of classes and scene contents in mask features, the amplitude components of mask features are replaced by a constant value of 1, which may lose the effective information of the mask features. Thus, when adding the proposed TMR module to the Earth-Adapter [30], the performance decreased. In conclusion, although adding the proposed TGDR approach to the Earth-Adapter [30] decreases the performance on the V2P benchmarks, the performance increased on the other three benchmarks, and the average performance of the four benchmarks improved by 0.8% mIoU compared with the Earth-Adapter [30]. This average performance improvement

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TABLE IV

COMPARISONS BETWEEN THE PROPOSED TGDR APPROACH AND RECENT DGSS METHODS ON THE P2V, V2P, R2U, AND U2R BENCHMARKS [26], [47]. THE "*" SYMBOL DENOTES THAT THE RESULTS ARE REPRODUCED BY OURSELVES

Methods	Venue	Avg	P2V	V2P	R2U	U2R
DAFormer [50]	CVPR'22	44.5	42.4	41.4	54.2	39.9
HRDA [51]	ECCV'22	39.6	33.1	31.1	54.2	39.8
FOSMix [34]	TGRS'24	40.9	51.9	43.2	34.9	33.7
CCDR [31]	TGRS'24	41.3	52.2	43.7	35.1	34.2
JSLS [32]	TCSVT'25	47.7	58.3	49.4	46.0	37.2
CrossEarth [35]	TPAMI'25	54.8	61.9	52.7	57.9	**46.6**
DINOv2+Mask2Former (Base) [19]	TMLR'24	53.1	62.2	50.5	55.9	43.9
+ Ours	-	55.1	64.5	53.3	58.1	44.6
Earth-Adapter* (Base) [30]	AAAI'26	54.7	61.6	54.5	58.3	44.4
+ Ours	-	55.5	63.1	54.2	58.8	45.8
Rein* (Base) [21]	CVPR'24	53.0	62.0	52.3	56.2	41.3
+ Ours	-	55.0	63.7	**54.7**	57.6	44.0
FADA* (Base) [17]	NIPS'24	53.5	60.6	52.4	57.0	43.9
+ Ours	-	55.1	63.5	53.9	58.3	44.6
SET* (Base) [29]	ACM MM'24	52.2	60.4	49.7	56.5	42.2
+ Ours	-	54.2	63.1	51.8	57.8	44.1
SoMA* (Base) [42]	CVPR'25	54.1	62.3	52.8	58.1	43.3
+ Ours	-	**56.0**	**65.6**	54.0	**59.0**	45.2

TABLE V

COMPARISONS BETWEEN THE PROPOSED TGDR APPROACH AND RECENT DGSS METHODS ON THE A2S AND S2A BENCHMARKS [49]. THE "*" SYMBOL DENOTES THAT THE RESULTS ARE REPRODUCED BY OURSELVES

Methods	Venue	Avg	A2S-Building	S2A-Building
DAFormer [50]	CVPR'22	35.7	7.1	64.3
HRDA [51]	ECCV'22	35.3	7.9	62.7
CrossEarth [35]	TPAMI'25	54.4	44.3	64.5
MID [48]	JSTARS'25	56.6	44.3	68.9
DINOv2+Mask2Former (Base) [19]	TMLR'24	55.6	42.9	68.2
+ Ours	-	58.8	43.5	74.1
Earth-Adapter* (Base) [30]	AAAI'26	55.8	36.5	75.0
+ Ours	-	56.7	37.8	75.5
Rein* (Base) [21]	CVPR'24	57.6	40.7	74.4
+ Ours	-	59.4	42.9	75.8
FADA* (Base) [17]	NIPS'24	58.4	42.5	74.2
+ Ours	-	59.4	43.3	75.5
SET* (Base) [29]	ACM MM'24	55.7	41.6	69.8
+ Ours	-	58.4	43.1	73.6
SoMA* (Base) [42]	CVPR'25	63.6	48.4	78.8
+ Ours	-	**64.1**	**48.8**	**79.3**

demonstrates that adding the proposed TGDR approach to the Earth-Adapter could further improve the generalization ability of the model. Furthermore, when adding the proposed TDR and TMR modules to the FADA and SET methods, the performance increased in succession. In particular, the FADA and SET methods are similar to the Earth-Adapter, which refines the features in the frequency domain. However, different from the Earth-Adapter, which uses learnable weights to combine the spatial, low-frequency, and high-frequency features, the FADA and SET use a set of learnable tokens to refine the low-frequency and high-frequency components, and then combine them uniformly (simply concatenating them together). Thus, using the proposed TMR module to further purify the phase components from the mask features to obtain the scene content information is positive for the semantic segmentation task.

Moreover, the qualitative results in Figs. 6 and 7 show the accurate segmentation results after using the proposed TDR and TMR modules in succession, especially in the classes of "low vegetation," "tree," and "clutter" on the P2V and V2P benchmarks, and in the classes of "agricultural," "building," and "barren" on the R2U and U2R benchmarks.

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TABLE VI
EFFECTIVENESS OF THE PROPOSED TGDR APPROACH ON THE P2V BENCHMARK [26]

Methods	mIoU	Potsdam to Vaihingen per-class IoU (%)					
		impervious surface	building	low vegetation	tree	car	clutter
DINOv2+Mask2Former (Base) [19]	62.2	71.3	85.5	53.8	70.9	57.9	33.7
+ Ours	64.5	75.0	87.1	57.8	71.2	58.2	37.4
Earth-Adapter (Base) [30]	61.6	71.0	87.9	51.0	68.6	56.0	35.2
+ Ours	63.1	75.1	84.2	55.1	72.4	53.7	38.3
Rein (Base) [21]	62.0	74.2	85.1	53.5	72.7	57.5	29.2
+ Ours	63.7	74.6	86.7	57.8	72.4	57.6	33.2
FADA (Base) [17]	60.6	72.1	84.7	52.5	71.5	57.1	25.4
+ Ours	63.5	73.8	86.9	57.1	73.2	57.8	32.1
SET (Base) [29]	60.4	72.4	85.8	52.2	69.2	56.8	25.8
+ Ours	63.1	72.0	85.1	54.4	71.8	59.4	35.8
SoMA (Base) [42]	62.3	74.4	88.5	56.9	73.9	59.3	20.7
+ Ours	65.6	76.0	88.8	60.4	73.2	59.6	35.7

TABLE VII
EFFECTIVENESS OF THE PROPOSED TGDR APPROACH ON THE V2P BENCHMARK [26]

Methods	mIoU	Vaihingen to Potsdam per-class IoU (%)					
		impervious surface	building	low vegetation	tree	car	clutter
DINOv2+Mask2Former (Base) [19]	50.5	66.2	78.6	55.5	31.5	68.6	2.6
+ Ours	53.3	68.9	83.0	57.5	39.1	69.2	2.0
Earth-Adapter (Base) [30]	54.5	67.9	84.5	56.7	41.3	74.1	2.3
+ Ours	54.2	69.8	85.5	57.9	34.4	75.5	2.2
Rein (Base) [21]	52.3	66.5	81.1	55.2	39.2	69.1	2.3
+ Ours	54.7	68.8	83.3	58.8	43.5	71.3	2.8
FADA (Base) [17]	52.4	67.7	82.1	57.3	36.8	66.7	3.7
+ Ours	53.9	68.2	83.0	58.9	40.3	70.4	2.7
SET (Base) [29]	49.7	66.9	78.8	55.6	28.9	65.3	2.6
+ Ours	51.8	66.8	81.5	58.2	31.7	68.5	4.1
SoMA (Base) [42]	52.8	69.7	82.4	58.9	28.2	75.2	2.4
+ Ours	54.0	70.6	82.5	59.7	33.7	75.0	2.4

In addition, we further analyze the effectiveness of the proposed TDR and TMR modules. In particular, for the proposed TDR module, the effectiveness of the interclass weights W_s and interclass similarity S_c is analyzed. First, the performance gain of using interclass weights W_s is analyzed, which can achieve an improvement of 0.6% mIoU in the average performance. In particular, "w/o W_s" indicates that the interclass similarity S_c is directly utilized to calibrate the class-specific features. Furthermore, we analyze the effectiveness of the calibration of the class-specific features using the interclass similarity S_c. The experimental results demonstrate that when removing the interclass similarity S_c to calibrate class-specific features, the performance decreased by 1.0% mIoU in the average performance.

For the proposed TMR module, the effectiveness of refining the phase component of mask features and using the vector with all one values to replace the amplitude component of mask features is verified. First, to verify the effectiveness of refining the phase component of mask features, we show the average performance of refining the amplitude component of mask features, which decreased by 0.7% mIoU. Second, we use the amplitude component and refined phase component to recompose the frequency representation of mask features, where the average performance decreased by 0.5% mIoU.

D. Comparisons With Recent DGSS Methods

The proposed TGDR approach is evaluated on multiple DG-RSSS benchmarks, including the P2V, V2P, R2U, U2R, A2S, and S2A benchmarks. The comparison results between the proposed TGDR approach and recent methods are shown in Table IV. In addition, the proposed approach is evaluated on the A2S and S2A benchmarks from the WHU dataset, which includes imagery from aerial and satellite sources, introducing significant discrepancies in resolution, spectral range, and data acquisition conditions. The comparison results are shown in Table V.

As shown in Table IV, the proposed TGDR approach achieves better average performance than recent methods, such as JSLS [32] and CrossEarth [35]. Compared with the baselines, including DINOv2 [19], Earth-Adapter [30], Rein [21],

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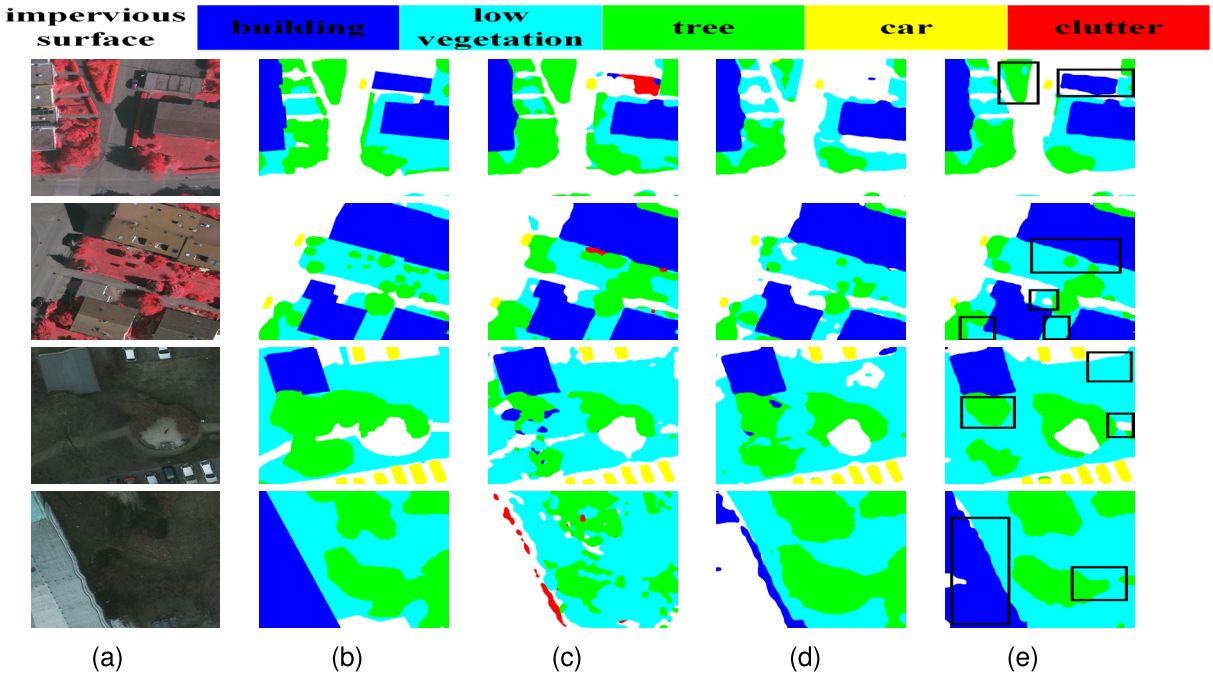


Fig. 8. Visualization of comparison on the P2V and V2P benchmarks [26]. The "EA" denotes the Earth-Adapter method. (a) Input. (b) GT. (c) Rein. (d) EA. (e) Ours.

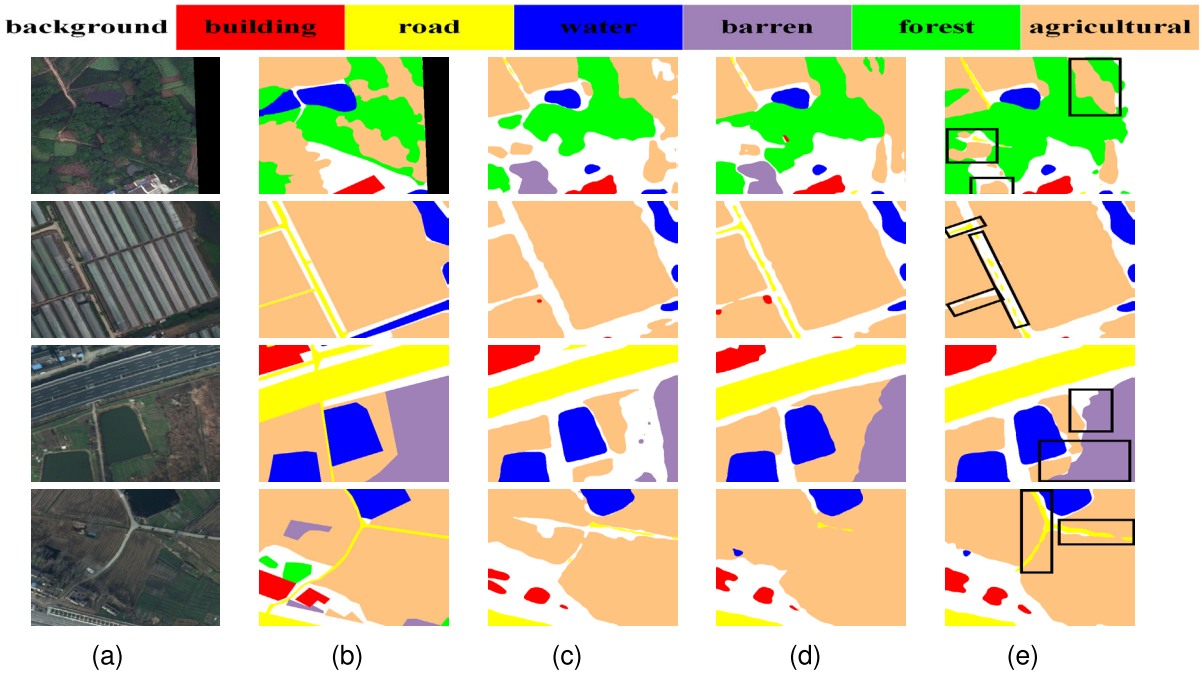


Fig. 9. Visualization of comparison on the R2U and U2R benchmarks [47]. The "EA" denotes the Earth-Adapter method. (a) Input. (b) GT. (c) Rein. (d) EA. (e) Ours.

FADA [17], SET [29], and SoMA [42], the proposed TGDR approach can achieve uniform gains, which improved by 2.0%, 0.8%, 2.0%, 1.6%, 2.0%, and 1.9% mIoU in the average performance. These performance improvements demonstrate that the proposed TGDR approach can be applied to these recent works to further improve the performance. In particular, the performances of each class are shown in Tables VI–IX. The proposed TGDR method can bring improvements to almost every category, especially in the classes of "impervious surface," "low vegetation," "tree," and "clutter" in the P2V and V2P benchmark, and the classes of "background," "building," "road," and "agricultural" in the R2U and U2R benchmarks. In addition, the visualizations shown in Figs. 8 and 9 also demonstrate that the proposed TGDR approach achieves accurate segmentation results in these classes.

As shown in Table V, the proposed TGDR approach also achieves reliable performance improvements in multiple baselines, including DINOv2 [19], Earth-Adapter [30], Rein

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TABLE VIII
EFFECTIVENESS OF THE PROPOSED TGDR APPROACH ON THE R2U BENCHMARK [47]

Methods	mIoU	Rural to Urban per-class IoU (%)						
		background	building	road	water	barren	forest	agricultural
DINOv2+Mask2Former (Base) [19]	55.9	38.1	65.5	53.3	73.2	49.2	56.7	55.4
+ Ours	58.1	41.3	65.4	54.7	75.6	50.5	56.6	62.3
Earth-Adapter (Base) [30]	58.3	41.7	66.6	55.7	75.9	52.8	56.0	59.5
+ Ours	58.8	43.2	65.9	56.5	75.7	52.3	56.1	62.0
Rein (Base) [21]	56.2	41.6	62.9	55.1	73.6	44.5	53.4	61.8
+ Ours	57.6	40.1	64.7	54.5	75.5	51.5	56.1	60.9
FADA (Base) [17]	57.0	40.4	65.4	54.8	74.5	46.5	56.3	61.1
+ Ours	58.3	41.1	67.4	55.0	75.7	50.9	57.4	60.3
SET (Base) [29]	56.5	39.9	65.9	54.9	74.5	45.7	55.5	59.3
+ Ours	57.8	40.8	65.2	55.3	75.6	50.2	56.6	60.9
SoMA (Base) [42]	58.1	42.9	66.2	55.6	75.3	48.4	55.8	62.5
+ Ours	59.0	43.3	63.3	56.9	76.0	53.3	54.8	65.5

TABLE IX
EFFECTIVENESS OF THE PROPOSED TGDR APPROACH ON THE U2R BENCHMARK [47]

Methods	mIoU	Urban to Rural per-class IoU (%)						
		background	building	road	water	barren	forest	agricultural
DINOv2+Mask2Former (Base) [19]	43.9	53.6	43.1	49.5	67.2	25.3	14.3	54.1
+ Ours	44.6	57.4	58.1	48.9	65.5	8.7	18.2	55.4
Earth-Adapter (Base) [30]	44.4	48.7	53.9	44.9	65.0	10.6	31.4	56.6
+ Ours	45.8	55.7	61.6	47.7	64.9	6.3	28.4	55.7
Rein (Base) [21]	41.3	51.8	58.9	47.2	62.8	5.8	13.6	48.9
+ Ours	44.0	57.6	60.8	47.0	63.1	7.8	21.7	50.0
FADA (Base) [17]	43.9	57.5	55.9	44.4	63.0	19.5	20.5	46.3
+ Ours	44.6	55.0	62.7	47.2	66.5	14.7	17.3	49.1
SET (Base) [29]	42.2	51.4	61.9	46.9	53.5	5.7	23.2	52.5
+ Ours	44.1	49.5	63.7	48.1	56.8	7.7	29.6	53.8
SoMA (Base) [42]	43.3	56.7	57.5	35.9	64.8	11.6	21.8	54.5
+ Ours	45.2	55.5	65.1	46.1	65.4	3.5	20.0	61.0

[21], FADA [17], SET [29], and SoMA [42]. In particular, the proposed TGDR approach achieves superior performance when it is applied to the SoMA method, which obtains 64.1% mIoU in the average performance between the A2S and S2A benchmarks. Compared with recent works, e.g., CrossEarth and MID, the proposed TGDR achieves performance improvements of 9.7% and 7.5% in terms of mIoU.

E. Analysis of Confusion Matrices

To verify the improvement of class discriminability in the proposed TDGR approach, the confusion matrices between the baseline and the proposed approach are analyzed, which are shown in Fig. 10. From these confusion matrices, compared with the "Base" and Earth-Adapter, after using the proposed TDGR approach, the average accuracy of predictions can achieve a promising improvement. For example, in the P2V benchmark, compared with the "Base," which achieves 80.3% average prediction accuracy for each class, after using the proposed TDGR approach, the average prediction accuracy achieves 83.6%, improved by 3.3%. In addition, in some similar classes, such as "low vegetation" and "tree," the misclassification has been alleviated.

F. Experiments on Learnable Tokens With Different Initialization Strategies

To further analyze the contribution of the initialization strategy adopted by the learnable token in the TDR module to the performance, we conduct comparative experiments on the learnable token with different initialization strategies. In particular, the initialization strategies include random initialization without using the CLIP model and class-related embeddings generated by using class-name prompt templates (e.g., "[cls]") and two different prompt templates, where one of the prompt templates is used in the proposed TDR module, that is, "a remote sensing photograph of [cls]." Another prompt template is "an urban scene photograph of [cls]."

As shown in Table X, first, compared with random initialization, although the performance gain in the average performance seems moderate (+0.4% mIoU), using "RS [cls]" prompt templates to generate text embeddings for

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Fig. 10. Visualization of confusion matrices. From top to bottom, each row represents the confusion matrix in P2V, V2P, R2U, and U2R benchmarks [26], [47], respectively. The "Base" denotes the DINOv2 with Mask2Former. The "EA" denotes the Earth-Adapter method. (a) Base. (b) Base+Our. (c) EA. (d) EA+Our.

TABLE X

EXPERIMENTS ON LEARNABLE TOKENS WITH DIFFERENT INITIALIZATION STRATEGIES ON THE P2V, V2P, R2U, AND U2R BENCHMARKS [26], [47]. IN PARTICULAR, THE "[CLS]", "US [CLS]", AND "RS [CLS]" INDICATE DIFFERENT PROMPT TEMPLATES, INCLUDING CLASS NAME, "AN URBAN SCENE PHOTO OF [CLS]", AND "A REMOTE SENSING PHOTO OF [CLS]"

Strategies	Average	P2V	V2P	R2U	U2R
Randomly Initialization	54.5	63.5	52.9	57.5	43.9
$[cls]$	54.2	63.2	52.4	57.8	43.2
US $[cls]$	53.4	62.8	51.3	57.7	41.8
RS $[cls]$	54.9	63.9	53.6	58.0	44.2

initializing the learnable tokens achieves steady performance improvements on P2V (+0.4% mIoU), V2P (+0.7% mIoU), R2U (+0.5% mIoU), and U2R (+0.3% mIoU), respectively. Second, generic or illmatched text prompts achieve suboptimal performance. We argue that the plain class name may lack contextual scene information to distinguish visually similar remote sensing categories, while the urban scene prompt may

TABLE XI

COMPARISON EXPERIMENTS OF THE PROPOSED TMR MODULE WITH DIFFERENT AMPLITUDE VALUES ON THE P2V, V2P, R2U, AND U2R BENCHMARKS [26], [47]

TMR	Average	P2V	V2P	R2U	U2R
M_f^{amp}=0	52.1	60.6	49.1	56.5	42.1
M_f^{amp}=0.5	53.0	62.2	50.4	57.1	42.4
M_f^{amp}=1.0	53.9	62.5	51.4	57.5	44.1
M_f^{amp}=3.0	53.9	62.4	51.2	57.5	44.3
M_f^{amp}=5.0	53.8	62.4	51.0	57.6	44.0
M_f^{amp}=10.0	53.5	62.3	51.2	57.5	42.9
M_f^{amp}=50.0	53.6	62.2	51.3	57.3	43.7
M_f^{amp}=100.0	53.3	61.9	51.1	57.1	43.2
M_f^{amp} (original amplitude)	53.4	62.3	51.1	56.4	43.8

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TABLE XII

ANALYSIS OF THE COMPUTATIONAL COST. ALL EXPERIMENTS IN THIS TABLE ARE CONDUCTED ON A SINGLE NVIDIA RTX 3090 GPU, WHERE THE BATCH SIZE IS SET TO 4, AND THE RESOLUTION OF THE INPUT IMAGE IS SET TO 512 × 512

Methods	FLOPs	Inference Latency	Total Parameters	Trainable Parameters	Training Memory Usage
Baseline [19]	310.85 GMac	16.3 iter/s	304.24 M	0 M	6966 M
+ TDR	336.98 GMac	15.2 iter/s	388.97 M	1.58 M	12070 M
+ TMR	310.88 GMac	15.8 iter/s	388.03 M	0.64 M	7510 M
+ Ours	336.98 GMac	14.8 iter/s	389.03 M	1.64 M	12854 M

introduce strong viewpoint bias (e.g., driving viewpoint) and cannot generalize to remote sensing imagery.

In conclusion, different initialization strategies have different contributions to the performance in the proposed TDR module, and the "a remote sensing photograph of [cls]" prompt template can bring better performance gain.

G. Comparison Experiments of TMR Module With Different Amplitude Values

We have further clarified the rationale of setting the amplitude components to the all-1 vectors by adding comparison experiments using different amplitude values in the proposed TMR module. In particular, first, as shown in Table XI, compared with using the original amplitude components and directly setting the amplitude components to 0, setting the amplitude components to a constant achieves better average performance. In particular, when directly setting the amplitude components to 0, the average performance decreased severely. We argue that when the amplitude components are set to 0 [$\hat{M}_f$ equals 0 in (13), that is, $\hat{M}_z$ has degenerated into M_z in (15)], inverse Fourier transformation cannot further mine scene contents from mask features. Second, as the amplitude increases from 0 to 5.0, the average performance first continues to rise and then stabilizes, reaching an average performance of 53.9% mIoU. Finally, when the fixed amplitude scalar exceeds 5.0, the average performance gradually degrades as the value grows larger. We argue that the decreased average performance is caused by excessive amplitude components, which may bring distribution noise.

In conclusion, from these experimental results, setting the amplitude components to 1.0–5.0 can eliminate domain-specific information within the amplitude components while maintaining the scene contents within the phase components, thereby achieving better average performance.

H. Analysis of Computational Cost

The computational cost of the proposed approach is further analyzed. As shown in Table XII, first, compared with the baseline, the computational cost of the proposed approach increased in the FLOPs, total parameters, and training memory usage, which is caused by the use of the CLIP model and matrix multiplication of the proposed TDR module. Second, compared with the baseline, the computational cost of the proposed approach achieves a minor increase in the trainable parameters, since the proposed approach only contains a set of learnable tokens with the dimensions of $L \times C \times M$, where $L = 24$, $C = 6$ on the P2V and V2P benchmarks or $C = 7$ on the R2U and U2R benchmarks, and $M = 1024$, and several linear projections, which introduce limited trainable parameters. Finally, compared with the baseline, the inference latency of the proposed approach has only a slight decrease, which may be mainly caused by the matrix multiplication.

In conclusion, although the proposed approach introduces the CLIP model and matrix multiplication during the training stage, leading to an increase in training memory usage, FLOPs, and total parameters, the inference latency and trainable parameters of the proposed approach increased slightly, and the performance increased significantly, which could be accepted.

TABLE XIII

EXPERIMENTS ON THE P2V, V2P, R2U, AND U2R BENCHMARKS [26], [47] UNDER DIFFERENT RANDOM SEEDS

Methods	Seed	Avg	P2V	V2P	R2U	U2R
Baseline [19]	42	53.1	62.2	50.5	55.9	43.9
	43	53.2	61.7	50.8	56.7	43.6
	44	53.0	62.1	50.5	55.8	43.4
+ Ours	42	55.1	64.5	53.3	58.1	44.6
	43	55.3	64.1	53.5	58.0	45.7
	44	55.4	64.6	54.4	57.8	44.8
SoMA [42]	42	54.1	62.3	52.8	58.1	43.3
	43	54.6	63.8	52.4	58.2	43.8
	44	54.3	62.6	53.2	57.9	43.4
+ Ours	42	56.0	65.6	54.0	59.0	45.2
	43	56.1	65.4	53.9	59.0	46.1
	44	56.3	65.1	54.8	58.7	46.5

I. Experiments on Different Random Seeds

We conduct repeated experiments with three distinct random seeds (42, 43, and 44) over all four cross-domain benchmarks, and the full quantitative results are summarized in Table XIII. First, the random fluctuations of average performance in the baselines are extremely limited. The average performance fluctuates within a range of less than 0.5% mIoU across three seeds, which verifies that the proposed approach is stable and reproducible. Second, compared with the baselines, the proposed TGDR achieves consistent positive improvements under all random seeds. For example, when combined with SoMA, the average performance consistently increases by 1.5%–2.0% under every seed setting. In conclusion, the minor variance across different random seeds fully demonstrates that the performance promotion brought by the proposed approach

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TABLE XIV
EXPERIMENTS WITH THE RARE CLASS SAMPLING STRATEGY ON THE U2R BENCHMARK [47]. THE "RCS" DENOTES THE RARE CLASS SAMPLING STRATEGY [50]

Methods	mIoU	Urban to Rural per-class IoU (%)						
		background	building	road	water	barren	forest	agricultural
SoMA (Base) [42]	43.3	56.7	57.5	35.9	64.8	11.6	21.8	54.5
+ RCS	43.9	52.8	61.9	42.8	60.1	14.1	19.7	56.1
+ Ours	45.2	55.5	65.1	46.1	65.4	3.5	20.0	61.0
+ RCS	45.7	51.1	63.5	46.7	63.6	10.6	25.6	59.0

is not an accidental result caused by specific parameter initialization, which demonstrates the effectiveness of the proposed approach.

V. LIMITATION AND FUTURE WORK

Although the proposed TGDR approach shows performance improvements in multiple DG-RSSS benchmarks, the proposed TGDR approach struggles with the segmentation in tail classes, such as the "clutter" in the P2V and V2P benchmarks, and "barren" and "forest" classes in the R2U and U2R benchmarks. We argue that the decreased performance in the tail classes may be due to the inevitable learning of a long-tail bias of classes during the fine-tuning stage, which is not only present in the proposed TGDR approach. As shown in the comparison experiments on the U2R benchmark (in Table IX), compared with DINOv2 [19]+Mask2Former [20], the recent methods, including Earth-Adapter [30], Rein [21], SET [29], FADA [17], and SoMA [42], show severe performance degradation on tail classes such as "barren" class. To alleviate this issue, we use the classic rare class sampling (RCS) strategy [50] during the fine-tuning stage. The comparison experiments are shown in Table XIV. After using the RCS strategy, the performance on the "barren" class increased by 2.5% and 7.1% in terms of mIoU, which demonstrates that increasing the sampling of tailed classes can improve the performance on the tailed classes to some extent. Thus, in future work, designing an effective PEFT strategy with tailed class sampling for improving the performance of tailed classes during the fine-tuning VFMs stage is a valuable research direction.

VI. CONCLUSION

In this article, a TGDR approach is proposed for DGSS in remote sensing. In particular, this work addresses inaccurate segmentation between similar classes and boundary regions using the proposed TDR module and the text-guided mask feature refinement module. The former uses class-related learnable tokens and interclass similarity to refine the original features from the frozen backbone for generating class-discriminated features. The latter relies on class-related learnable tokens to inject class-specific semantics into the phase component of mask features for refining the prediction of class boundaries. Extensive ablation and comparison experiments demonstrate the effectiveness of the proposed two modules and the superior performance of the proposed approach on multiple DG-RSSS benchmarks.

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